Day 8 - Joule Agent Designs (designs.md)

Warm-up (1 h): 3 agent designs using the AGENT DESIGN TEMPLATE. Risky step marked in **RED** = human must approve.

#	Agent	Auto when	Human approves when (red)
1	Requisition-to-PO Agent	Supplier is good AND amount <= Rs 25 lakh -> create PO.	**Supplier is new or poor, OR amount > Rs 25 lakh.**
2	Order Credit-Check Agent	Exposure <= credit limit AND no payment overdue > 30 days -> release order.	**Exposure > limit, OR overdue > 30 days, OR new customer, OR order value > Rs 10 lakh.**
3	Leave Request Agent	Balance enough AND <= 3 days AND team coverage >= 70% AND no blackout date -> approve.	**Balance short, OR > 3 days, OR clash with team absence / blackout, OR sensitive type (medical, maternity, unpaid).**

Design 1 - P2P Requisition Agent

AGENT NAME	Requisition-to-PO Agent
GOAL (1 line)	Turn an approved purchase requisition into a PO, or route it to a human when risky.
STARTS WHEN	A new requisition is created or released.
SKILLS (3-5)	1. check_supplier - look up supplier status (good / new / poor) 2. check_amount - compare with the Rs 25 lakh limit 3. create_po - write to po_created.json (never a real system) 4. route_to_manager - send to procurement manager with reason 5. write_log - log decision + reason
AUTO WHEN	Supplier is good AND amount <= Rs 25 lakh -> create PO.
HUMAN WHEN	**Supplier is new or poor, OR amount > Rs 25 lakh.**
ESCALATE TO	Procurement manager (no reply in 2 working days -> Head of Procurement).
LOG THESE	requisition id, supplier, amount, supplier status, rule applied, decision, reason, timestamp.

Checklist: 3-5 small skills [x] | at least one human approval [x] | clear trigger [x] | log list [x]

Design 2 - O2C Credit Check Agent

AGENT NAME	Order Credit-Check Agent
GOAL (1 line)	Decide if a sales order can be released or must be held for credit review.
STARTS WHEN	A sales order is saved and flagged by the credit check.
SKILLS (3-5)	1. get_customer_credit - credit limit, open items, overdue invoices 2. calc_exposure - open orders + open invoices + this order vs limit 3. check_payment_history - anything overdue > 30 days? 4. release_order - remove credit block (simulated: released_orders.json) 5. route_to_credit_manager - hold order and send with reason
AUTO WHEN	Exposure <= credit limit AND no payment overdue > 30 days -> release order.
HUMAN WHEN	**Exposure > limit, OR overdue > 30 days, OR new customer, OR order value > Rs 10 lakh.**
ESCALATE TO	Credit manager (Finance); above Rs 50 lakh -> Finance Controller + Sales Head.
LOG THESE	order id, customer, limit, exposure, overdue days, decision, reason, approver, timestamp.

Checklist: 3-5 small skills [x] | at least one human approval [x] | clear trigger [x] | log list [x]

Design 3 - Leave Request Agent

AGENT NAME	Leave Request Agent
GOAL (1 line)	Approve simple leave requests automatically and send tricky ones to the manager.
STARTS WHEN	An employee submits a leave request.
SKILLS (3-5)	1. check_leave_balance - enough days left? 2. check_team_calendar - how many teammates already off? 3. check_blackout_dates - year-end close, go-live, audit weeks 4. approve_leave - record approval and update balance 5. route_to_manager - send with reason, notify employee
AUTO WHEN	Balance enough AND <= 3 days AND team coverage >= 70% AND no blackout date -> approve.

HUMAN WHEN	**Balance short, OR > 3 days, OR clash with team absence / blackout, OR sensitive type (medical, maternity, unpaid).**
ESCALATE TO	Line manager; no action in 2 days -> HR partner.
LOG THESE	request id, employee id, dates, days, balance before/after, coverage %, decision, reason, approver, timestamp. (No medical details.)

Checklist: 3-5 small skills [x] | at least one human approval [x] | clear trigger [x] | log list [x]

Partner Review Sheet

To be completed by the reviewing partner (they must find at least one risk per design).

Design	Risk found / missed	Reviewer name	Date	Result
1 P2P Requisition				
2 O2C Credit Check				
3 Leave Request				